Module 4 — Synthesis Quiz

Question bank · After exercise lab · open notes · two attempts

Module 4, Synthesis quiz

Readings: Read 0, Read 1, Read 2
Labs: Demo, Exercise
Canvas: Synthesis quiz (Module 4) · **two attempts**, highest score counts

Full integration for Project 4: interpret two-group CIs, connect study design to causal language, read `t.test()` / `prop.test()` output, simulation workflow, and **four lab situations** (numerical vs binary outcome; validate against known truth).

Instructions

- Open notes (methods map, R reference).
- Use the CI interpretation template from Read 2: *"I am 95% confident that…"*
- For R items, match the workflow in the demo lab and your Project 4 templates.

Section A — Interpret two-group confidence intervals

Use: *"I am 95% confident that [population parameter difference] is between [L] and [U]. In this context, confidence means that if data from many samples were collected the same way, then 95% of the intervals computed would contain the true difference."*

A1. Sleep intervention CI

RCT; 95% CI for mean sleep change (treatment − control): [**0.4, 1.8**] hours. Interpret in context.

A2. Recovery proportion CI

95% CI for difference in recovery proportions (new − standard treatment): [**0.05, 0.22**]. Interpret.

A3. Zero in the CI

95% CI for mean score difference (tutoring − control): [**$-$2.1, 8.4**]. Interpret. What does zero in the interval mean?

A4. Spot incorrect interpretation

Which is the **best** interpretation of a 95% CI for a mean difference?

A) 95% probability the difference is in this specific interval.
B) Repeated sampling: ~95% of such intervals contain the true parameter.
C) 95% of individuals differ by this amount.

Answer key — Section A — Interpret two-group confidence intervals

1. I am 95% confident the true mean increase in sleep for the treatment group compared to control is between 0.4 and 1.8 hours (with confidence-level sentence).
2. I am 95% confident the true difference in recovery proportions (new minus standard) is between 0.05 and 0.22 — the new treatment has a higher recovery rate in the population.
3. Zero is a plausible value → we **cannot** conclude a nonzero population difference at the 95% level. We are 95% confident the difference is between $-$2.1 and 8.4 points.
4. **B** is correct. A treats the parameter as random after data are observed; C confuses parameter with individual observations.

Section B — Study-design integration (causal vs association)

B1. Causal conclusion in RCT

RCT: 120 volunteers randomly assigned to two diets; weight loss compared after 12 weeks. p = 0.01. Write a valid causal conclusion.

B2. Association-only conclusion

Observational: employees who use standing desks report 15% higher productivity. p = 0.04. Write a valid (non-causal) conclusion.

B3. Identify causal overreach

A news headline reads: "Coffee **causes** longer life — study of 50,000 adults." The study was observational. What is wrong with the headline?

B4. When RCT is impossible

Name one situation where an RCT is impossible or unethical and explain why observational methods are used instead.

Answer key — Section B — Study-design integration (causal vs association)

1. **Valid:** Random assignment supports a causal interpretation — the low-carb diet caused greater weight loss on average (p = 0.01), ruling out confounding by baseline differences.
2. **Valid (association-only):** Employees who use standing desks are associated with higher productivity (p = 0.04), but unmeasured confounders (e.g. health motivation) could explain the difference.
3. The study is **observational** — causation cannot be established because coffee drinkers may differ from non-drinkers in unmeasured ways (e.g. diet, socioeconomic status). The headline overstates the design's conclusions.
4. Examples: studying the effect of **smoking** on lung cancer (cannot assign people to smoke); **natural disasters** on mental health; **childhood poverty** on outcomes. Observational methods are used because random assignment is unethical or infeasible.

Section C — R workflow for two-group comparison (Read 2 · Lab)

These R patterns support the simulation and analysis in the demo lab and Project 4.

C1. set.seed()

What does `set.seed(123)` do and why is it used in simulation?

C2. group_by + summarize

Write a tidyverse pipeline that computes mean and SD of `outcome` for each level of `group` in `df`.

C3. boxplot by group

Write a ggplot2 call for a **boxplot** of `outcome` by `group` in `df`.

C4. t.test formula interface

Which R call runs a Welch two-sample t-test of `outcome` by `group` in `df`?

C5. prop.test two groups

Two groups: group A has $x_1 = 30$ successes in $n_1 = 80$; group B has $x_2 = 45$ in $n_2 = 90$. Write the `prop.test` call.

C6. stacked bar

Write ggplot2 code for a **100% stacked bar** comparing binary outcome proportions across two groups.

Answer key — Section C — R workflow for two-group comparison (Read 2 · Lab)

1. Fixes the **random number generator state** so that simulation results are **reproducible** — running the script again gives the same random draws.
2. `df %>% group_by(group) %>% summarize(mean = mean(outcome), sd = sd(outcome))`
3. `ggplot(df, aes(x = group, y = outcome)) + geom_boxplot()`
4. `t.test(outcome ~ group, data = df)` — formula interface; Welch is the default (`var.equal = FALSE`).
5. `prop.test(c(30, 45), c(80, 90))`
6. `ggplot(df, aes(x = group, fill = outcome)) + geom_bar(position = 'fill') + labs(y = 'Proportion')`

Section D — Error types in two-group design context

D1. Drug trial errors

Test if a new drug reduces symptoms more than placebo ($H_a : \mu_{\text{drug}} < \mu_{\text{placebo}}$). State Type I and Type II errors in context.

D2. Training program errors

Test if a training program improves pass rates ($H_a : p_{\text{trained}} > p_{\text{control}}$). State errors in context.

D3. Decision and error type

Welch t-test; $p = 0.08$; $\alpha = 0.05$. Which decision? Which error type is now possible?

D4. Decision and error type 2

Two-proportion z-test; $p = 0.003$; $\alpha = 0.05$. Decision? Which error type is possible?

Answer key — Section D — Error types in two-group design context

1. **Type I:** conclude drug is better when it is not (approve ineffective drug). **Type II:** fail to detect that the drug works (miss effective treatment).
2. **Type I:** conclude training helps when it does not. **Type II:** miss a beneficial training program.
3. **Fail to reject H_0.** **Type II** error is now possible — we might be missing a real difference.
4. **Reject H_0.** **Type I** error is possible — we might be incorrectly concluding the proportions differ.

Section E — Demo lab: simulation + two-group test situations

Match each scenario to the correct test and simulation plan. See the demo lab Problems 1–4.

E1. Problem 1 — Simulate and test numerical outcome

Binary EV (treatment vs control), numerical RV (e.g. recovery days). $n_1 = 30$, $n_2 = 30$, true $\mu_1 = 10$, $\mu_2 = 12$.
Which simulation approach and which test?

E2. Problem 2 — Simulate and test binary outcome

Binary EV, binary RV (success/fail). $n_1 = 50$, $n_2 = 50$, true $p_1 = 0.4$, $p_2 = 0.6$.
Simulation and test?

E3. Problem 3 — Validate simulation

After simulating with a known true difference, how do you **validate** that your code is correct?

E4. Problem 4 — RCT interpretation

Simulated RCT with random assignment: p = 0.02 for group difference. Write the conclusion connecting the design to causation.

Answer key — Section E — Demo lab: simulation + two-group test situations

1. Simulate with `rnorm(30, mean = 10)` and `rnorm(30, mean = 12)`. Test: **Welch two-sample t-test** (`t.test(days ~ group)`).
2. Simulate with `rbinom(50, 1, 0.4)` and `rbinom(50, 1, 0.6)`. Test: **two-proportion z-test** (`prop.test(c(x1, x2), c(50, 50))`).
3. Run many simulations (e.g. 1000 reps); check that **the estimated rejection rate near the true effect** matches expected power, and that the CI covers the true parameter ~95% of the time.
4. Because of **random assignment**, the group difference is attributable to the simulated treatment, not confounding. We reject H_0 (p = 0.02) and conclude the treatment caused the observed difference in the simulation.
